Laboratory of Biomedical Optics – Biophotonics

Exercise 3: Inverse Problem - Optical Properties Retrieval by Non-Linear Fitting

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1 Introduction

In Exercises 1 and 2, we focused on the *forward problem*: given known optical properties (μ_a, μ_s', n) and geometry, predict the time-resolved photon distribution $\Phi(\mathbf{r}, t)$.

In clinical and preclinical biomedical diagnostics (e.g., non-invasive tissue oximetry, functional brain imaging, optical mammography), we face the *inverse problem*: given experimental measurements of light escaping from the tissue, reconstruct the unknown optical properties $\mathbf{p} = [\mu_a, \mu_s']$.

In this laboratory we will use Monte Carlo data as our "experimental" data, and we will try to retrieve the optical parameters knowing the source-detector separation.

Forward Model Recap

In Exercise 2, we derived the time-resolved surface flux $J_z(\rho, t)$ for a semi-infinite medium with Extrapolated Boundary Conditions (EBC). The analytical solution is:

$$J_z(\rho, t) = \frac{v}{2A}(4\pi D v t)^{-3/2} \left[\exp\left(-\frac{\rho_+^2}{4Dvt}\right) - \exp\left(-\frac{\rho_-^2}{4Dvt}\right)\right] \exp(-\mu_a v t) \tag{1}$$

where $D \approx \frac{1}{3\mu_s'}$, $v = c_0/n$, and ρ_+^2 is the positive source-detector distance and ρ_-^2 is the negative source-detector one. As already done in the previous laboratory, the helper functions `Afactor.m` and `TDSemiInfinite_EBC_normalized.m` are provided.

Curve Normalization

In practical time-correlated single-photon counting (TCSPC) systems, measuring absolute photon numbers requires complex radiometric calibration and suffers from unknown optical fiber coupling losses and detector quantum efficiency.

To make optical properties estimation robust against unknown scaling factors, both the measured curve $y_{\text{data}}(t)$ and the analytical forward model $y_{\text{model}}(t)$ are normalized to unit area:

$$\tilde{y}(t) = \frac{y(t)}{\int_0^\infty y(t')\,dt'} \approx \frac{y(t_i)}{\sum_i y(t_i)} \tag{2}$$

The inverse problem then relies exclusively on the shape of the Distribution of Time of Flight (DTOF).

Non-Linear Least Squares Optimization

The inverse problem is formulated as finding the parameter vector $\mathbf{p}^* = [\mu_a, \mu_s']$ that minimizes the sum of squared residuals:

$$\min_{\mu_a, \mu_s'} S(\mu_a, \mu_s') = \sum_{i=1}^{N} \left[y_{\text{data}}(t_i) - y_{\text{model}}(t_i; \mu_a, \mu_s')\right]^2 \tag{3}$$

We will use MATLAB's `lsqcurvefit`.

To quantify fit quality, we evaluate the residual sum of squares (unweighted χ^2):

$$\chi^2 = \sum_{i=1}^{N} \left[y_{\text{data}}(t_i) - y_{\text{model}}(t_i; \mu_a, \mu_s') \right]^2 \tag{4}$$

2 Objectives

1. Implement non-linear curve fitting in MATLAB using `lsqcurvefit` to solve the inverse problem.
2. Retrieve optical properties from benchmark Monte Carlo data (`Homo_MC.mat`) across different source-detector separations ($\rho = 1, 2, 3, 4\,\text{cm}$).
3. Investigate the sensitivity of the inversion to the initial parameter guesses ($[\mu_{a0}, \mu_{s0}']$).
4. Implement the χ^2 metric and compare the fit error with the error evaluated at the exact Monte Carlo simulation parameters ($\mu_a = 0.1\,\text{cm}^{-1}$, $\mu_s' = 10\,\text{cm}^{-1}$).
5. Understand the fundamental distinction between measurement noise and systematic forward-model approximation error. Does minimizing χ^2 with a Diffusion model recover the true physical optical properties when the data is generated by exact transport physics (Monte Carlo)?

3 Step-by-Step Instructions

Step 1: Environment and Data Loading

Define the experimental temporal grid, load the simulated data, and normalize it to unit area.

```
clear; close all;

% Physical and temporal parameters
n = 1.4;                    % refractive index of the medium
dt = 5;                     % time step [ps]
N = 2000;                   % number of time points
t = (0:N-1)' * dt + dt/2; % time vector [ps]

% Load simulated Monte Carlo data
load Homo_MC.mat;           % contains yMC (2000x4)
% iMC = 1 -> SD = 1 cm, iMC = 2 -> SD = 2 cm, etc.
iMC = 3;                    % select distance (e.g. 3 cm)
ydata = yMC(:, iMC);

% Define source and detector positions
SourcePos = [0, 0, 0];
DetPos = [iMC, 0, 0];       % [cm]

% Normalize data to unit area
ydata = ydata ./ sum(ydata);
```

Step 2: Define Initial Guess and Model Function

Define an initial guess for the unknown parameters and visualize the initial model vs. the data.

Step 3: Setup and Execute Non-Linear Fitting

Use the function `lsqcurvefit` to retrieve $[\mu_a, \mu_s']$. Refer to MATLAB's guide. Use the following function options:

```
options = optimoptions('lsqcurvefit', 'Display', 'iter-detailed', 'PlotFcns'
    , @optimplotfval);
```

Try to to fit the 4 curves at different source-detector distance. What do you observe? Try different set of initial parameters and fit. What do you observe as ρ increases? Is the result changing with the initial guess? How does the number of iterations change with the distance of the initial guess from the solution?

Step 4: χ^2 Calculation

Implement the calculation of the unweighted χ^2.

1. For $\rho = 3\,\text{cm}$, compute χ^2_{true} using the *exact parameters* of the Monte Carlo simulation ($\mu_a = 0.1\,\text{cm}^{-1}$, $\mu_s' = 10\,\text{cm}^{-1}$):

2. Compute χ^2_{fit} using the *fitted parameters* ($\mu_a^{\text{fit}}, \mu_s'^{\text{fit}}$). Compare χ^2_{true} and χ^2_{fit}. Which value is smaller? Why do the fitted parameters give a *better* fit ($\chi^2_{\text{fit}} < \chi^2_{\text{true}}$) to the Monte Carlo curve than the true physical parameters that were actually used to generate the data? Explain the difference between *stochastic measurement noise* and *systematic forward model error* (DE vs. RTE).

4 Extra Questions

1. Why is area normalization necessary when fitting experimental TCSPC data in practice? What optical information is lost by normalizing to unit area, and what information is preserved?

2. If experimental noise is Poisson-distributed ($\sigma_i^2 \approx y_{\text{data}}(t_i)$), how should the objective function in weighted least squares be modified? Which parts of the DTOF curve (peak vs. tail) receive higher relative weight?

3. Why does cross-talk between μ_a and μ_s' occur, and how does increasing the temporal measurement dynamic range (e.g., 3–4 decades of decay) help decouple them?

4. In a clinical fNIRS experiment measuring brain oxygenation through the intact scalp and skull, why would a single-distance homogeneous diffusion fit yield an "effective" or "apparent" optical property rather than true brain tissue values?